10(a)	energy = $mc\Delta T$	**C1**
	energy = ItV $(\Delta T =) \dfrac{0.40 \times 0.020 \times 75\,000 \times 0.95}{0.015 \times 130}$	**C1**
	$=290$ K	**A1**
10(b)	$I = I_o e^{-\mu t}$ $0.20 = e^{-0.22t}$	**C1**
	$t = 7.3$ cm	**A1**

Question	Answer	Marks
10(c)	*either* (linear) attenuation coefficients / μ very different for bone and muscle	**M1**
	(very) different amounts (of X-rays) absorbed so good contrast or (very) different intensities transmitted so good contrast	**A1**
	or (linear) attenuation coefficients / μ similar for blood and muscle	**(M1)**
	similar amounts (of X-rays) absorbed so poor contrast or similar intensities transmitted so poor contrast	**(A1)**

Question	Answer	Marks